U-Net with a Pretrained ResNet-50 Encoder, Multi-Annotator Training and Morphological Instance Extraction

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. We treat the task as semantic segmentation of the union of annotated filaments followed by a simple instance extraction. A U-Net with an ImageNet-pretrained ResNet-50 encoder is trained at 1024^2 resolution on a randomly chosen annotator record per image and epoch, with re-estimated batch-normalisation statistics, and is combined with flip test-time augmentation and a morphology-based post-processing. The final model, trained on all 707 labelled images, reaches a public leaderboard PQ of 0.34; the same configuration trained without the validation images reaches 0.44 on our date-grouped validation split. We report ablations, an uncertainty analysis, and an error analysis showing that the remaining errors are dominated by objects whose overlap with the annotation is close to the IoU threshold, and by small filaments.

CCS Concepts

• **Computing methodologies** → **Image segmentation**.

Keywords

solar filaments, instance segmentation, U-Net, ResNet, panoptic quality, multi-annotator labels

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1]. Our solution follows three principles: (i) use a strong pretrained encoder, because only 707 labelled images are available; (ii) train on what the metric rewards, i.e. on the records of all annotators; and (iii) keep the instance extraction simple, so that every step can be verified against the organizers' scoring code. Design decisions were made on a date-grouped validation split and checked against the public leaderboard with a small number of controlled submissions. Code, the trained weights, and a notebook that reproduces the submitted predictions from the stored weights are provided in our repository (https://github.com/parsaBehjatTabrizi/solar-filament-segmentation-2026).

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026
2026.

2 Methodology

Data and sampling strategy. The training set contains 707 GONG H-α images (2048×2048, grayscale) with 1 154 annotation records; a record is the set of polygons drawn by one annotator on one image, so an image can have several records. We split the 707 images into 568 training and 139 validation images, grouping by observation date so that no date appears in both parts (random 80/20 split of the dates, seed 42). The validation images carry 225 annotator records. The 180 test images were never used for any decision. For the final model we retrain with the same configuration on all 707 images; no held-out score exists for it, so its public leaderboard score is its only external check. Test images are, if anything, closer in time to training images than our validation images (median gap to the nearest training image 2 days for both sets; 17 % of test images share a date with a training image, none of the validation images), so date-grouped validation is not optimistic with respect to temporal proximity.

Model. We use a U-Net [9] whose encoder is a ResNet-50 [4] initialised from the public torchvision ImageNet weights [3]. The one-channel image is repeated to three channels and ImageNet-normalised inside the network. The five encoder stages (strides 2–32) feed a decoder of five double-convolution blocks (conv–BN–ReLU twice) with bilinear up-sampling and skip connections, followed by a 1×1 convolution; the network has 32.6 M parameters and outputs a per-pixel filament probability.

Training. Images are resized to 1024^2 (area interpolation) and normalised by their own 1st–99th intensity percentiles; the only augmentation is horizontal and vertical flipping. The target is the union of the polygons of *one* annotator record, drawn uniformly at random among the image's records at every epoch. Because the official metric scores every record separately, this trains on what the metric rewards and averages out annotator-specific choices. We minimise binary cross-entropy plus soft Dice loss [8] with Adam [6] (decoder learning rate 10^{-3}, encoder 10^{-4}, weight decay 10^{-5}), cosine decay to 1 % over 50 epochs, batch size 4 and mixed precision. Our first models showed erratic validation behaviour (validation-loss spikes and one epoch with PQ 0.000); re-estimating the batch-normalisation running statistics [5] on 240 training images restored PQ 0.27 for that checkpoint, so we re-estimate them before every

validation and for the final weights (the learned weights are unchanged). With this fix the last epoch is as good as the best one, so no checkpoint selection is needed for the final model.

Inference and post-processing. A test image is resized and normalised as above and the network is applied to the image and to its horizontally, vertically and doubly flipped versions; the four probability maps are averaged. The map is up-sampled bilinearly to 2048^2 and thresholded at 0.5; a morphological closing with an elliptical kernel of radius 10 px re-joins fragments of one filament; 8-connected components become the instances, and components of at most 400 px are discarded. Each instance is written as a run-length encoded mask (Figure 1). The threshold, the minimum size and the closing radius were chosen on the validation split (grid: threshold 0.4–0.6, size 200–800 px, radius 0–15 px), using a split-half check over validation dates so that a setting was only adopted if it helped on both halves.

3 Evaluation

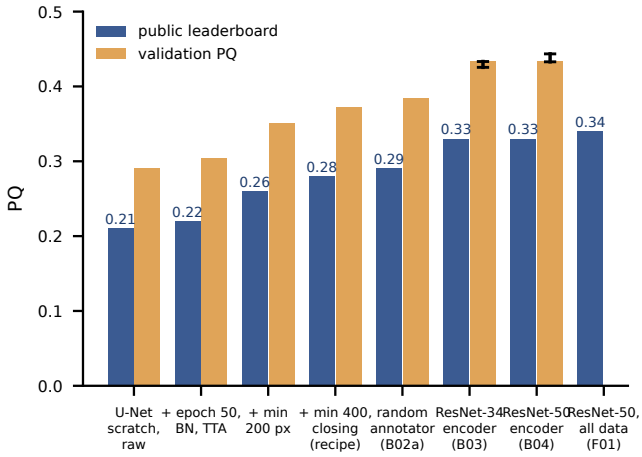


Figure 3: Public leaderboard PQ (blue) and validation PQ (orange) of the successive configurations; the first four use the same plain U-Net trained from scratch. Black bars show the range of two independent trainings of the same configuration. Intermediate validation values use the primary-record protocol (within 0.002 of the official protocol where both were computed).

Strategy. We use the organizers' metric: pooled panoptic quality [7] with many-to-many matching, i.e. every predicted/ground-truth pair with IoU > 0.5 is a true positive and contributes its IoU, unmatched predictions are false positives, unmatched ground-truth instances are false negatives, and $\mathrm{PQ} = \sum \mathrm{IoU}/(\mathrm{TP} + \frac{1}{2}\mathrm{FP} + \frac{1}{2}\mathrm{FN})$. The predictions of an image are scored separately against *every* annotator record and the counts are pooled. Our implementation reproduces the organizers' scoring code to six decimals on validation predictions written in the submission format. Validation was reused to choose post-processing settings and to compare models, so validation PQ is slightly optimistic; the public leaderboard (a subset of the test set) is the independent check.

Results. Figure 3 summarises the development. The leaderboard score is consistently about 0.09–0.10 below validation, but every change moved both almost equally, so validation is a reliable guide to relative improvement. Post-processing contributed the largest gain (validation 0.29→0.37, leaderboard 0.21→0.28), the pretrained encoder the second (0.38→0.43; 0.29→0.33), and training on all images raised the leaderboard score from 0.33 to 0.34. A deeper encoder (ResNet-50 vs. ResNet-34), averaging both models (0.439 vs. 0.4435 for the best single model) and instance-level confidence filtering (+0.005, on one half of the dates only) gave no reliable gain. Retraining the same configuration changed validation PQ by up to 0.011 (ResNet-34: 0.4333 vs. 0.4255; ResNet-50: 0.4329 vs. 0.4435), so differences below about 0.01 are within training noise. A paired bootstrap over observation dates (2 000 resamples, all records of an image kept together) gives a 95 % interval of $[0.42, 0.46]$ around the validation PQ of 0.444 of the ResNet-50 model trained without the validation images, and a gain of +0.053 (95 % interval $[+0.043, +0.064]$) of the full post-processing over thresholding alone, which was positive in all resamples.

Strengths, weaknesses and possible improvements. On validation, the ResNet-50 model trained without the validation images has TP = 1 063, FP = 577 and FN = 544, a mean IoU of matched pairs of 0.677 and a detection quality $\mathrm{TP}/(\mathrm{TP} + \frac{1}{2}\mathrm{FP} + \frac{1}{2}\mathrm{FN})$ of 0.655 (PQ = 0.4435). Passing the exact ground-truth union through our resolution, closing and size filter gives PQ = 0.83, so instance extraction is not the main limit. Most errors are borderline: 365 of the 577 false positives overlap an annotated filament with IoU ≤ 0.5 (212 touch no annotation), and 220 of the 544 misses overlap exactly one prediction with IoU ≤ 0.5 (243 have no overlapping prediction). Only 56 misses would be repaired by merging fragments, and 35 predictions merge two annotated filaments. Small filaments (200–999 px) are hardest: 49 % of the predictions are false positives and 42 % of the annotations are missed. Pooled PQ by station ranges from 0.41 to 0.47 and by year block from 0.43 to 0.47. Annotators also disagree: on the 59 validation images with several records, another annotator's record scores 0.38 PQ against the primary record under the same metric, so we regard annotator agreement as a reference level, not as a ceiling for models. The method is simple, stable and reproducible, but it cannot separate touching filaments (it predicts a semantic mask), has no explicit boundary model, and its hyper-parameters were tuned on a single validation split. Promising directions are boundary-aware or probabilistic multi-annotator targets, higher input resolution for small filaments, and cross-validation over date blocks. The leaderboard result (0.34) is below the validation estimate, and we could not attribute this gap to temporal distance; other candidates are label differences between the hidden test annotations and ours, and the small size of the public subset.

Reproducibility and tools. The final weights are stored in half precision (65 MB) in the repository; the reproduction notebook loads them and regenerates the submitted file with an agreement (pooled PQ between the two files) of 0.9994. The solution was developed in PyTorch on Kaggle GPUs; parts of the code and of this report were written with the assistance of a large language model, and all methodological choices and results were reviewed by the author.

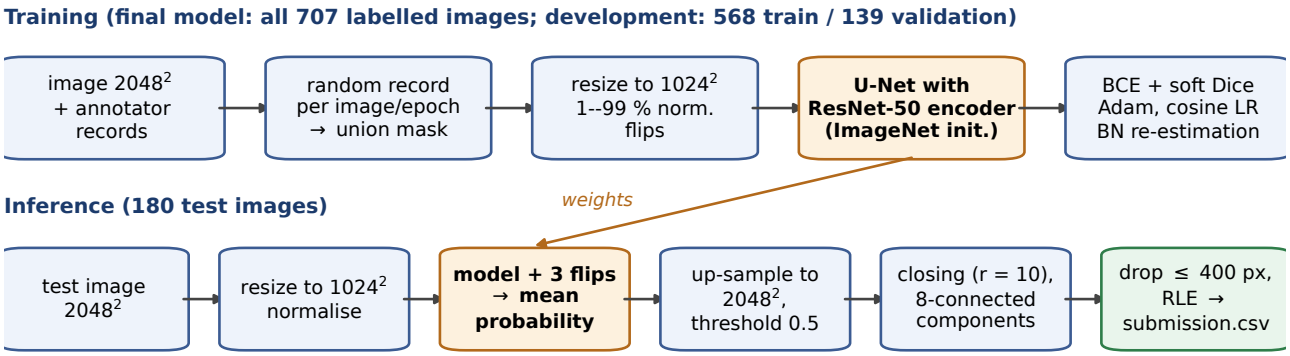


Figure 1: End-to-end pipeline. Top: training. Bottom: inference on a test image; the orange boxes mark the trained network.

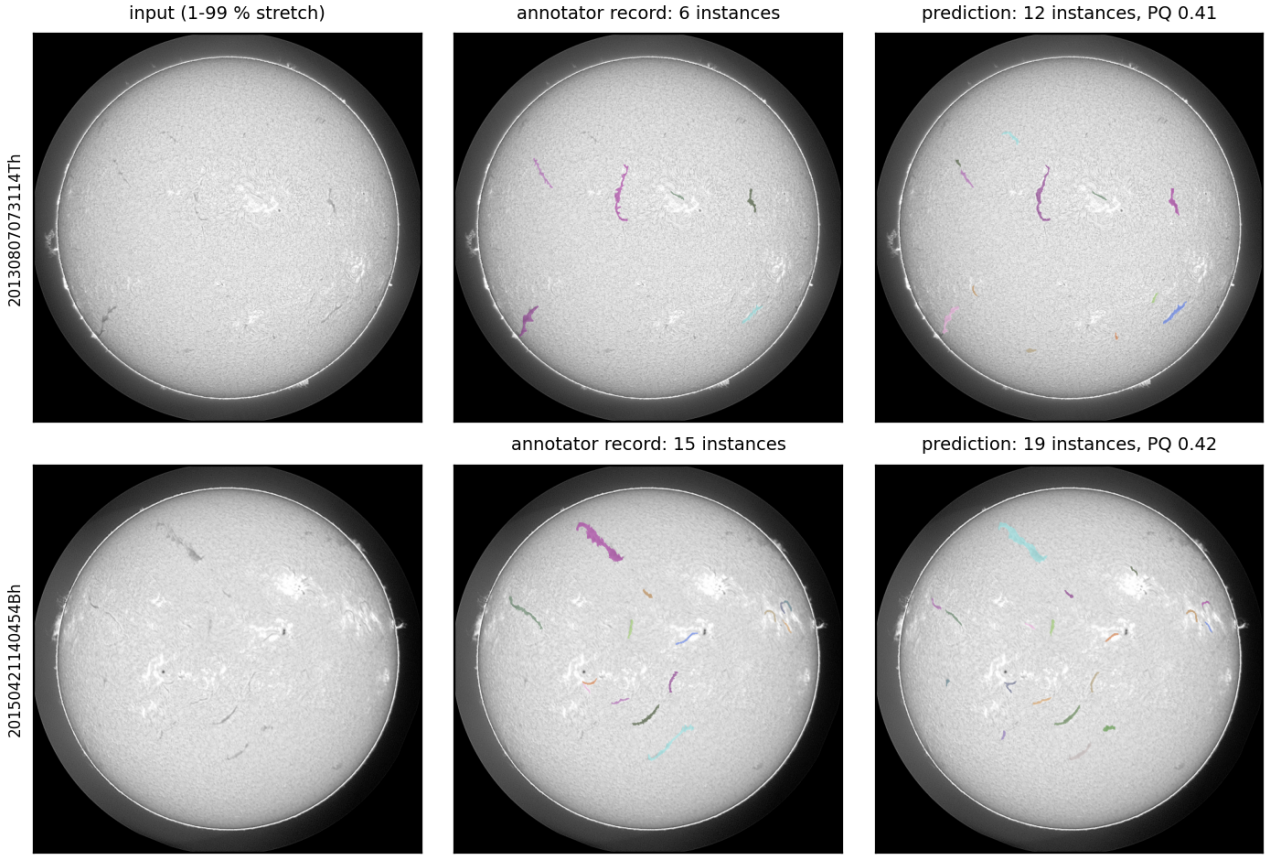


Figure 2: Segmentation examples on validation images that were not used for training (two images drawn at random from different stations, not selected): input, instances of one annotator's record, and our predicted instances (colours only distinguish instances; the PQ shown is pooled over all records of the image).

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